

Reproduction of meta-pytorch/torch tune: Qwen2.5-0.5B LoRA fine-tuning recipe

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Independent reproduction · Other · Natural language processing

Abstract

Ran the lora_finetune_single_device recipe with the shipped config qwen2_5/0.5B_lora_single_device at the pinned commit: LoRA (rank 32) fine-tuning of Qwen2.5-0.5B-Instruct on alpaca-cleaned for one epoch, bf16, batch 2 with 8 accumulation steps, seed 42; only the model and output paths were overridden. With the config's default learning rate (2e-3, no gradient clipping) the loss falls until the end of warmup at step 100, then diverges and does not recover. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Qwen2.5-0.5B LoRA, default ...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: Qwen2.5-0.5B LoRA, default recipe

Run	Metric	Final value	Step
1	train_loss	7.4928	3235
1	steps	3235	
1	peak_memory_reserved_gb	5.32812	
1	mean_tokens_per_second	3028.81	

Run	Inline claim	Value	Phase
1	first_step_loss	1.89042	
1	final_loss_mean_last100	7.22038	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	3392.7 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	ec1d407ccace9678ff591fb0cf31d60555516588063d57ad98646753982a1f2b	7.2 MB
r1.kvbundle.zip	cf2c2c93a72bd28c8f756591cfc27cae1df2addacd52a2e4d13c8669c4680ec4	2.0 MB

5 Links and licence

Reproduction of: <https://github.com/meta-pytorch/torch tune/tree/bd2a0fc7c31430972728494fa01aaeb0ebf1ba1>

Record: CC-BY 4.0. Code: BSD-3-Clause, as declared by the authors. Tags: lora, fine-tuning, qwen2.5, llm.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Qwen2.5-0.5B LoRA, default recipe), kv:2610.00017v1/r1. Data hash 83e379ad2c69e034a106600a4146911d1837488067750279186839bc636e7a3b.

<https://kveritas.org/records/2610.00017v1/r1.pdf>